

AE-MoSE: an AutoEncoder Mixture of Spatial Experts for Geodemographic Classification

Stef De Sabbata^{*}, Pengyuan Liu^{**}, Owen Goodwin^{***}, and Alex Singleton^{***}

^{*}School of Geography, Geology and the Environment, University of Leicester, UK

^{**}Urban Analytics Subject Group, Urban Studies & Social Policy Division, University of Glasgow, UK

^{***}School of Environmental Sciences, University of Liverpool, UK

Abstract

This work introduces a novel approach to geodemographic classification that combines an AutoEncoder architecture with a Mixture of Experts framework, incorporating a Graph Neural Network into the gating component to create a Mixture of Spatial Experts approach (AE-MoSE). We provide a description of the approach and illustrate some preliminary results.

Keywords: Geodemographics, AutoEncoder, Mixture of Experts, Graph Neural Network

1. Introduction

Geodemographic classifications have largely relied on centroid-based clustering methods since their inception (Webber and Craig 1976, 1978). *K-means* remains a central approach to creating geodemographic classifications to this day (Wyszomierski et al. 2024), while most spatial geodemographic classification approaches are based on *fuzzy c-means* (Grekousis 2021), which enables fuzzy assignment of areas to classes rather than exclusive assignment, while also accounting for the spatial aspect of the data. These clustering techniques are often combined with dimensionality reduction methods such as principal component analysis (Liu, Singleton, and Arribas-Bel 2019), to preprocess the data and produce values suitable for *k-means* clustering. De Sabbata and Liu (2019, 2023) developed the use of autoencoders as a more effective dimensionality reduction approach to capture the underlying manifold (Buchanan et al. 2025) and to account for its spatial aspects. Nevertheless, *k-means* continues to serve as the foundation of geodemographic classification.

In this extended abstract, we propose an AutoEncoder Mixture of Spatial Experts approach (AE-MoSE) to replace the use of *k-means* and enable the creation of geodemographic classifications through end-to-end training of a deep learning model. An examination of our preliminary results suggests that AE-MoSE is a robust approach, able to capture key aspects of the 2021 UK Census.

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2. Methods

AE-MoSE combines the autoencoder architecture (Bourlard and Kamp 1988) with the mixture of experts framework (Jacobs et al. 1991) for unsupervised classification (Kopf et al. 2021). We propose a model defined as a combination of an encoder, a set of P expert decoders and two components defining a two-level hierarchical gating structure. In general, any of the encoder, decoder and gating networks can be constructed as a simple Multi-Layer Perceptron (MLP) or a spatially-explicit neural network, using graph neural network layers (De Sabbata and Liu 2023) or location encoding (Mai et al. 2022), thus varying the quantitative and qualitative attention that the overall model pays to the spatial aspects of the data (Dryden and Hoefler 2022; Wang, Yang, and Che 2025). In the following, we present a model that adopts a spatially-explicit neural network as the first-level gating component, while using simple a-spatial MLPs for the encoder, the expert decoders and the second-level gating component¹.

Each output area is represented by a vector $\mathbf{x} \in \mathbb{R}^N$ containing N variables derived from the 2021 UK Census and normalised in the range $[0 \dots 1]$. The encoder uses an MLP to generate the embeddings $\mathbf{z} \in \mathbb{R}^Z$ from $\mathbf{x}$, with $Z \ll N$. The first level of the gating network estimates the assignment $\mathbf{g}'$ of each census output area over the P' groups of expert decoders based on the input $\mathbf{x}$, using a GraphSAGE (Hamilton, Ying, and Leskovec 2018) layer, which takes into account the geographic context of each output area, followed by an a-spatial MLP. The second level of the gating network consists of one MLP per group, each processing the embeddings $\mathbf{z}$ to produce a within-group assignment $\mathbf{g}''$. The two are then combined to estimate the final assignment $\mathbf{g}$ for each output area over the P experts as $\mathbf{g}(p', p) = \mathbf{g}'(p') \cdot \mathbf{g}''(p', p)$, for each group p' and expert p . Both assignments use sampling from the Gumbel-Softmax distribution. Each decoder expert is an MLP tasked with creating a reconstruction $\mathbf{x}' \in \mathbb{R}^N$ of the original input from the embeddings $\mathbf{z}$.

The loss is defined as the autoencoder reconstruction loss combined with two auxiliary loss components. The first component is the balance loss defined by Shazeer et al. (2017), to minimise the variation in the importance of each expert, where the latter is calculated as the sum of its assignment values. The second component is an expert specialisation loss, which penalises the gating network for being too uncertain in assigning routing probabilities, based on the entropy of the assignment values. Both auxiliary loss components are applied at the group level (on $\mathbf{g}'$) and at the combined expert level (on $\mathbf{g}$).

We trained a model on 171 variables, adopting a mirror encoder-decoder design with layers of size 192, 128, 96, and 64, with $P = 21$ expert decoders. The first-level gating component uses a GraphSAGE layer with 128 outputs, using a neighbours graph $\mathcal{N}$ based on queen’s adjacency and eight-nearest neighbours, followed by an MLP with sizes 64, 32 and 7, generating the assignments $\mathbf{g}'$ to $P' = 7$ groups. The second-level gating component is composed of $P' = 7$ corresponding MLPs with sizes 32, 16 and 3, thus generating the final assignments $\mathbf{g}$ to the $P = 7 \times 3 = 21$ expert decoders. The assignments $\mathbf{g}$ are interpreted as 21 geodemographic classes, aiming to create a classification akin to the 21 groups of the 2021 Output Area Classification (2021 OAC, Wyszomierski et al. 2024).

1. Code available at: <https://github.com/stefdesabbata/aemose-geoai26>

The model was trained for 1500 epochs using *AdamW* on a random sample of 80% of the 239,023 UK output areas, with batches of 512 output areas, retaining 10% for validation and 10% for testing. We set the contributions of the balancing loss components to 0.05 with a 0.1 tolerance for the group routing, and 0.1 with a 0.1 tolerance for the expert routing, and the contribution of the specialisation loss components to 0.005 for both group and experts. The temperature of the Gumbel-Softmax distribution was annealed from 5.0 to 1.0 over 300 epochs starting at epoch 30, and switching to hard routing for groups at the epoch 500.

3. Preliminary results

Figure 1 illustrates the results obtained when applying the best-performing checkpoint (at epoch 1381) of the model described in the previous section to the entire dataset. Experts in Groups 4 and 6 capture the UK countryside, while experts in the other groups focus on the urban and suburban areas, and experts frequently show correspondence with classes in the 2021 OAC. We found no substantial overfitting during training, as the train, validation, and test sets yield similar values, indicating that the model is capable of generalising the classification to unseen data.

To provide a preliminary extrinsic assessment of the proposed approach, we explored how representative the 21 AE-MoSE expert classes are of crimes reported in 2025 in England and Wales² (see, e.g., Gale, Singleton, and Longley 2015), compared to the 21 group classes of the 2021 OAC. For the assessment, we use the average distance per category to the ten closest crimes for each output area boundary. Table 1 shows that estimates based on the average per AE-MoSE expert perform better than those based on the groups (and frequently, marginally better than those based on the more granular 52 subgroups) of the 2021 OAC, suggesting that the classification is similarly capable of capturing socio-economic patterns. Moreover, we compared the ridge regression estimates from the model embeddings with those from the original model input values. The former performs commensurately with the latter, although with an expected underperformance, indicating that the embeddings capture meaningful patterns and retain a substantial fraction of the signal present in the input data.

Overall, these results show that the proposed AE-MoSE approach is capable of capturing key aspects of the UK’s socio-demographics, and suggest that the architecture might, when applied to broader datasets, be capable of producing general-purpose embeddings and support the development of a population dynamics foundation model (Agarwal et al. 2024). However, a thorough exploration of the hyperparameter space, a robust extrinsic assessment and the integration of explainability (Liu, Zhang, and Biljecki 2024) and interpretability (De Sabbata, Mizzaro, and Roitero 2025) approaches will be necessary to further validate the use of AE-MoSE for geodemographic classification.

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2. Data available at: <https://data.police.uk/data/archive/>

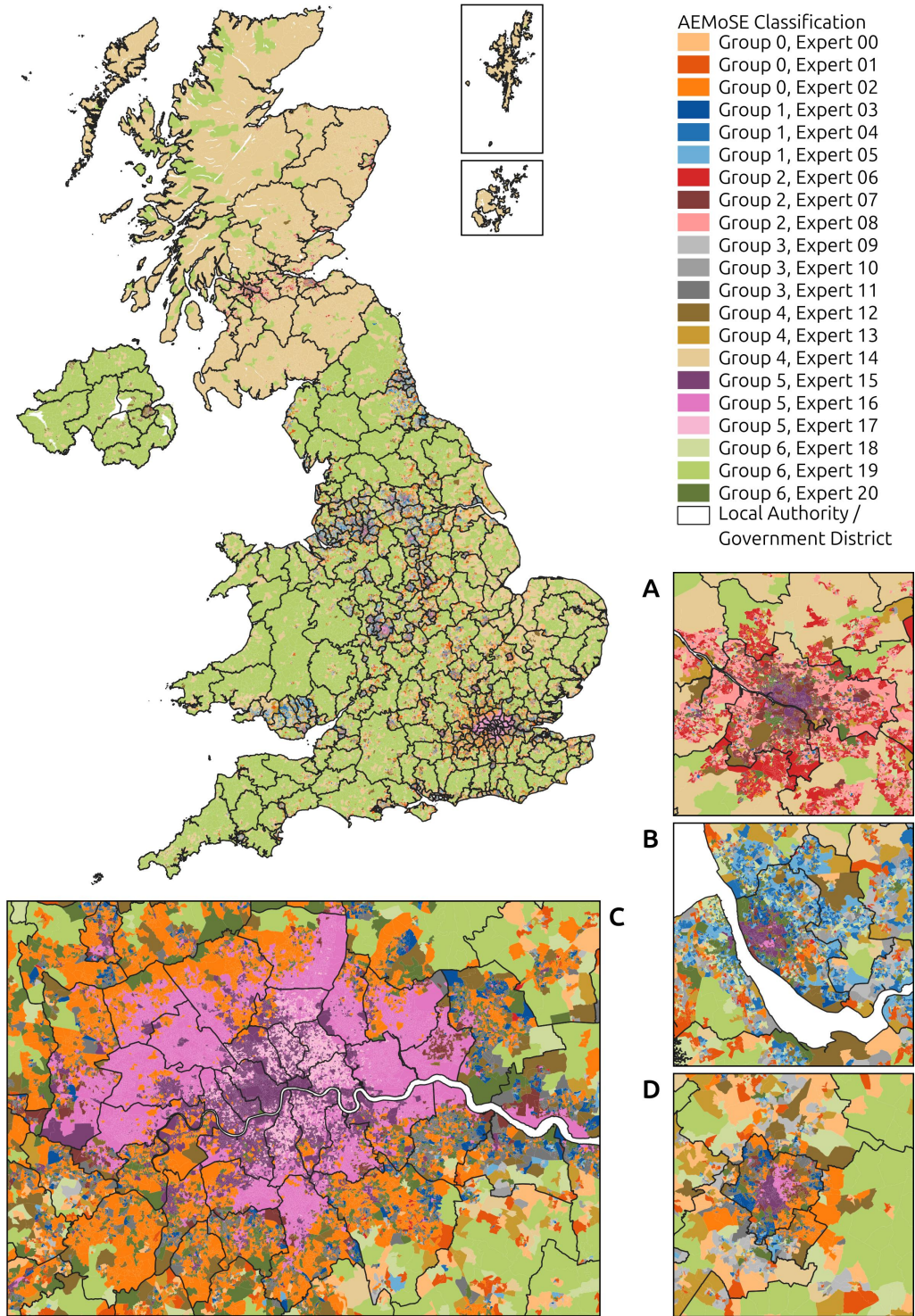


Figure 1: Map of the UK illustrating the classification obtained when applying the best-epoch model checkpoint on the whole dataset. The inset maps in the lower part of the image show the detail for Glasgow (A), Liverpool (B), London (C) and Leicester (D). Geometries source: Office for National Statistics, National Records of Scotland and Northern Ireland Statistics and Research Agency, licensed under the Open Government Licence v.3.0. Contains NRS, OSNI, and OS data © Crown copyright and database right 2023.

Category	AEMoSE exp	OAC grp	OAC sub	OAC sup	AEMoSE emb	input var
Anti-social behav.	0.195	0.177	0.198	0.157	0.274	0.299
Bicycle theft	0.403	0.374	0.399	0.343	0.506	0.551
Burglary	0.265	0.235	0.253	0.216	0.339	0.373
CD&A	0.191	0.184	0.202	0.167	0.310	0.342
Drugs	0.339	0.310	0.336	0.289	0.438	0.466
Other crime	0.223	0.194	0.212	0.166	0.343	0.377
Other theft	0.277	0.252	0.272	0.235	0.373	0.401
Poss. of weapons	0.277	0.253	0.272	0.228	0.401	0.438
Public order	0.220	0.201	0.219	0.180	0.345	0.369
Robbery	0.437	0.400	0.424	0.372	0.527	0.562
Shoplifting	0.173	0.143	0.160	0.109	0.272	0.298
Theft from the pers.	0.548	0.498	0.528	0.473	0.634	0.668
Vehicle crime	0.323	0.291	0.311	0.267	0.395	0.428
VSO	0.152	0.165	0.181	0.156	0.261	0.287

Table 1: R^2 values by crime category (left column), comparing estimations based on average by AE-MoSE expert and 2021 OAC classes (four central columns), and ridge regressions based on AE-MoSE embeddings and input census variables (two right columns). Targets were $\log(1 + d)$ -transformed before fitting.

References

- Agarwal, Mohit, Mimi Sun, Chaitanya Kamath, Arbaaz Muslim, Prithul Sarker, Joydeep Paul, Hector Yee, Marcin Sieniek, Kim Jablonski, Swapnil Vispute, et al. 2024. “General geospatial inference with a population dynamics foundation model.” *arXiv preprint arXiv:2411.07207*.
- Bourlard, Hervé, and Yves Kamp. 1988. “Auto-association by multilayer perceptrons and singular value decomposition.” *Biological cybernetics* 59 (4): 291–294.
- Buchanan, Sam, Druv Pai, Peng Wang, and Yi Ma. 2025. *Learning Deep Representations of Data Distributions*. Online, August.
- De Sabbata, Stef, and Pengyuan Liu. 2019. “Deep learning geodemographics with autoencoders and geographic convolution.” In *Proceedings of the 22nd AGILE conference on Geographic Information Science, Limassol, Greece*.
- . 2023. “A graph neural network framework for spatial geodemographic classification.” *International Journal of Geographical Information Science* 37 (12): 2464–2486. <https://doi.org/10.1080/13658816.2023.2254382>.
- De Sabbata, Stef, Stefano Mizzaro, and Kevin Roitero. 2025. *Geospatial Mechanistic Interpretability of Large Language Models*. arXiv: 2505.03368 [cs.LG].
- Dryden, Nikoli, and Torsten Hoefler. 2022. “Spatial Mixture-of-Experts.” In *Advances in Neural Information Processing Systems*, edited by S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, 35:11697–11713. Curran Associates, Inc.

- Gale, C, A Singleton, and P Longley. 2015. "Profiling burglary in london using geodemographics." In *Proceedings of the 23rd GISRUK conference*.
- Grekousis, George. 2021. "Local fuzzy geographically weighted clustering: a new method for geodemographic segmentation." *International Journal of Geographical Information Science* 35 (1): 152–174. <https://doi.org/10.1080/13658816.2020.1808221>.
- Hamilton, William L., Rex Ying, and Jure Leskovec. 2018. *Inductive Representation Learning on Large Graphs*. arXiv: 1706.02216 [cs.LG].
- Jacobs, Robert A., Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. 1991. "Adaptive Mixtures of Local Experts." *Neural Computation* 3 (1): 79–87. <https://doi.org/10.1162/neco.1991.3.1.79>.
- Kopf, Andreas, Vincent Fortuin, Vignesh Ram Somnath, and Manfred Claassen. 2021. "Mixture-of-Experts Variational Autoencoder for clustering and generating from similarity-based representations on single cell data" [in en], edited by Qing Nie. *PLOS Computational Biology* 17, no. 6 (June): e1009086. ISSN: 1553-7358. <https://doi.org/10.1371/journal.pcbi.1009086>.
- Liu, Pengyuan, Yan Zhang, and Filip Biljecki. 2024. "Explainable spatially explicit geospatial artificial intelligence in urban analytics." *Environment and Planning B: Urban Analytics and City Science* 51 (5): 1104–1123. <https://doi.org/10.1177/23998083231204689>.
- Liu, Yunzhe, Alex Singleton, and Daniel Arribas-Bel. 2019. "A Principal Component Analysis (PCA)-based framework for automated variable selection in geodemographic classification." *Geo-spatial Information Science* 22 (4): 251–264. <https://doi.org/10.1080/10095020.2019.1621549>.
- Mai, Gengchen, Krzysztof Janowicz, Yingjie Hu, Song Gao, Bo Yan, Rui Zhu, Ling Cai, and Ni Lao. 2022. "A review of location encoding for GeoAI: methods and applications." *International Journal of Geographical Information Science* 36 (4): 639–673. <https://doi.org/10.1080/13658816.2021.2004602>.
- Shazeer, Noam, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc V. Le, Geoffrey E. Hinton, and Jeff Dean. 2017. "Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer." *CoRR* abs/1701.06538.
- Wang, Yimeng, Zhiyao Yang, and Xiangjiu Che. 2025. "A Hierarchical Mixture-Of-Experts Framework for Few Labeled Node Classification." *Neural Networks* 188:107285. ISSN: 0893-6080. <https://doi.org/https://doi.org/10.1016/j.neunet.2025.107285>.
- Webber, Richard, and John Craig. 1976. "Which local authorities are alike." *Population Trends* 5:13–19.
- . 1978. *Socio-economic classification of local authority areas*. 35. HM Stationery Office.
- Wyszomierski, Jakub, Paul A. Longley, Alex D. Singleton, Christopher Gale, and Oliver O'Brien. 2024. "A neighbourhood Output Area Classification from the 2021 and 2022 UK censuses." E12550 GJ-AR-Jun-2023-0082.R1, *The Geographical Journal* 190 (2): e12550. <https://doi.org/https://doi.org/10.1111/geoj.12550>.